

NAPCORE Core Concepts

Introduction to semantic modelling

A **concept scheme** (in SKOS: *ConceptScheme*) is a structured collection of concepts that carry a fixed, shared meaning within a particular domain. The goal is for everyone working with the model — people and systems alike — to use the same words for the same things. That sounds obvious, but in practice language causes a surprising amount of confusion: what one organisation calls a *measurement point*, another calls a *sensor*, a *location*, or an *observation station*.

A concept scheme solves this by recording, for each concept:

- a **preferred label** (prefLabel) it is known by,
- a **definition** that unambiguously describes what the concept does and does not cover,
- and optionally a **scope note** or **example** to clarify borderline cases.

Concepts rarely stand on their own. They are related **hierarchically** (a vehicle is broader than a truck) or **associatively** (a road user relates to a road segment). These relationships make it possible for queries and analyses to navigate between general and specific concepts, and to make connections between concepts explicit. Within SKOS we use `broader`, `narrower`, and `related` for that.

When a concept also exists elsewhere — in a national or European standard, for instance — we record that as a **mapping relation**. This prevents reinventing the wheel: we link to the source so interoperability with external systems remains possible.

This document describes the concept scheme as it is currently recorded. The following sections list the metadata, every concept, the relationships between them, and — where the model is connected to a data source — how it is linked to that source data.

NAPCORE Core Concepts

Description

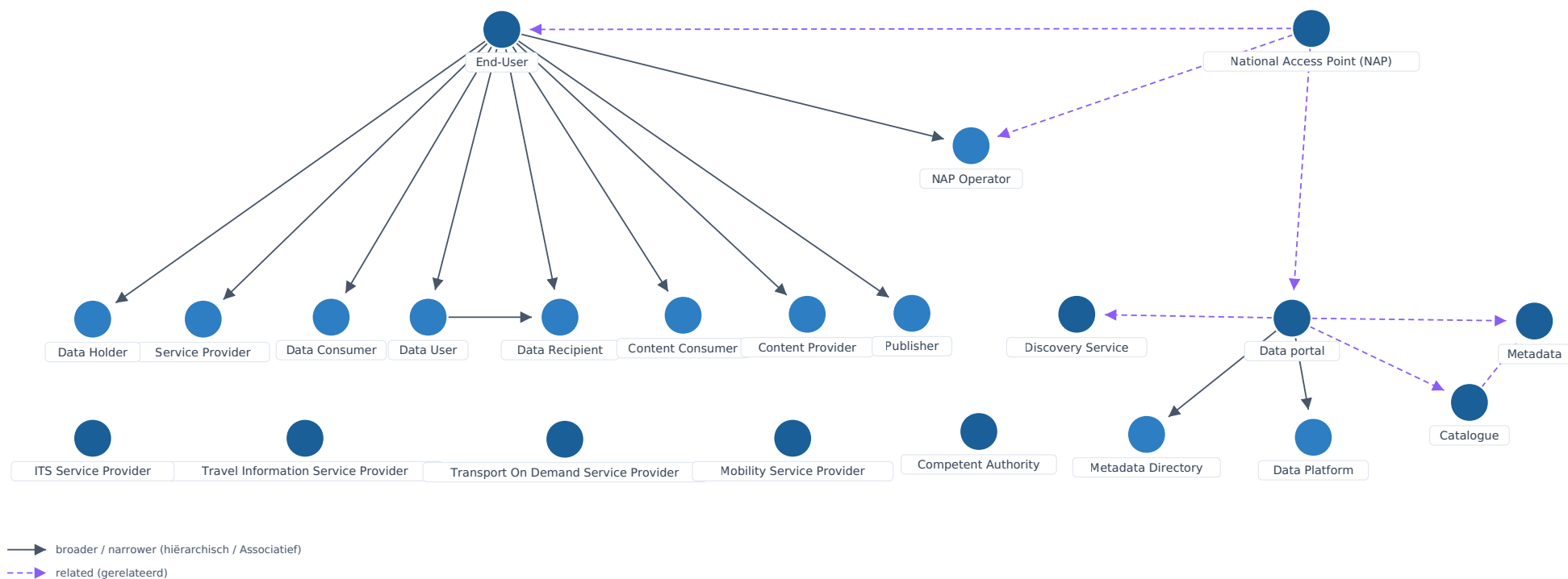
Core terminology of the NAPCORE reference architecture for National Access Points, covering NAP types, ecosystem roles and key mobility service concepts.

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Model overview

Visual overview of the concepts and their relations. Hierarchical relations are drawn with solid arrows; associative relations are drawn with dashed arrows.



Bron: Ontologica — NAPCORE Core Concepts

Concepts

Concepts overview

Concept	Definition	Alternative labels
National Access Point (NAP)	A National Access Point is a digital interface set up by a Member State to provide single-point access to data, in compliance with the Delegated Regulations of the European Commission that establish functional, technical, organisational, and service provisions to guarantee compatibility, interoperability, and continuity in the deployment and operational use of Intelligent Transport Systems. The NAPs organise access to and reuse of transport-related data to help support the provision of EU-wide interoperable travel and traffic ITS services to end users.	Data Portal
Data portal	Web-based system that contains a data catalogue with descriptions of datasets and provides services enabling the discovery and re-use of the datasets. In the context of mobility, a synonym to a data portal is a National Access Point (NAP), which is a node in which ITS-related data are concentrated and published in the form of datasets. It is set up to facilitate access, easy exchange and reuse of transport related data in Europe, in order to help support the provision of EU-wide interoperable travel and traffic services to end users.	
Catalogue	A concrete mobility data portal, i.e., a web portal, where mobility-related datasets and their distributions are described via metadata and discoverable for data users, e.g., the National Access Point web portal.	
Metadata Directory	A Metadata Directory is a National Access Point that contains descriptions of the data and services essential for their identification, assessment, and subscription, i.e., Metadata. It provides facilities to authenticate Data Providers and functionality to insert Metadata records. The Metadata Directory offers search and Discovery Services of the hosted records to end users and, through machine readable content, to other portals. The Metadata Directory is not directly involved in the data exchange between Data Providers and Data Consumers.	
Data Platform	A Data Platform is a National Access Point that contains descriptions of the data and services important for their identification, assessment, and subscription by, i.e., Metadata, together with the data and/or services themselves. It provides facilities to authenticate Data Providers, and optionally, Data Consumers, and allows data and metadata records to be inserted into the NAP's internal database. A Data Platform provides search and Discovery Services for the hosted records to End Users and, via machine readable content, to other portals and distribution interfaces. The Data Platform is an intermediary between Data Provider and Data Consumer, however, the data remain under the ownership of the Data Provider.	
Competent Authority	A Competent Authority / National Body is an organisation designated by a Member State to monitor the operation of the National Access Point and to verify compliance with the requirements established by the ITS Directive and its Delegated Regulations, including the prescribed	National Body

Concept	Definition	Alternative labels
	availability, format, and quality requirements for data and services made accessible through the National Access Point. Note: The term National Body can be used interchangeably with Competent Authority, but it carries an additional requirement of impartiality and independence for the organisation towards Service Providers, public and private Road Operators, Parking Operators and Broadcasters. The term remains valid in current legislation.	
NAP Operator	Public or private organisation designated to manage, maintain and operate the National Access Point, ensuring the accessibility of data and that the NAP functions in line with the requirements of the ITS Directive and its Delegated Regulations, further specified in national legislation or governance documents. Note: Usually, a department of the Member State's ministry responsible for transport, a state-owned organisation, or a research institution acts as NAP Operator. The organisation fulfilling this role is generally designated explicitly in national legislation or related governance documents	
Publisher	An entity (organisation) responsible for making the Catalogue available. This organisation is the direct contact for platform users, who have questions or issues about the platform and its system.	
Content Provider	Organisation or entity that supplies content (e.g., data, metadata, documents) to the National Access Point, making it available for reuse in accordance with the requirements of the ITS Directive and its Delegated Regulations. Note: Depending on the context, similar terms, more explicit or specific may be used: Data Holder, Data Provider, Service Provider, to describe the same role of supplying content to the National Access Point. Some examples of Content Providers: public and private road operators, broadcasters dedicated to traffic information, parking operators, road authorities, digital map producers, transport authorities, passenger transport operators, infrastructure managers, transport-on-demand service providers, travel information service providers.	
Data Holder	Any legal person, public or private entity, such as transport authorities, transport operators, infrastructure managers or transport on demand service providers, who has the right to grant access to or to share the data under its control.	
Service Provider	Public or private provider of a service. Specific interpretations of the term vary across EU legislation, with specialised definitions set out in the Delegated Regulations adopted under the ITS Directive and in the Regulation on the deployment of alternative fuels infrastructure.	
ITS Service Provider	Provider of an ITS service, whether public or private.	

Any public or private provider of at least one travel and traffic information, to Data Users and End Users, excluding a mere converter of information.

Concept	Definition	Alternative labels
Travel Information Service Provider		
Transport On Demand Service Provider	Any public or private provider of a real-time traffic information service, excluding a mere conveyer of data to data users.	
Mobility Service Provider	Legal person that provides services in return for remuneration to an End Users, including the selling of recharging or refuelling services.	
Content Consumer	Organisation or entity that accesses, retrieves, or uses data or information made available through the National Access Point for the purpose of developing services, applications, or analyses.	
Data Consumer	Entity, system, or application that consumes only the data made available through the National Access Point, such as datasets, APIs, or other machine-readable data, typically for technical, analytical, or operational processing.	
Data User	Any public or private entity, such as transport authorities, transport operators, Service Providers and infrastructure managers, research or non-governmental organisations, to create multimodal travel information, create services or perform research or analysis, or where provided by the terms and conditions determined by the data holder, using the data for other purposes.	
Data Recipient	Natural or legal person that has the right to access the data produced and made available by the Data Holder under specified access conditions for use.	
End-User	Any natural or legal person who receives, accesses or benefits from transport-related services, such as travel information services, road safety-related traffic information services, or alternative fuels infrastructure services.	
Metadata	Structured description of the contents of the data facilitating the discovery and use of the data.	
Discovery Service	Service allowing for the search of the requested data using the contents of the corresponding Metadata and displaying such contents.	